THESIS DISCUSSION · PROF. PRENDINGER

M2 Target Comparison

Triple-barrier labels · four backends · 20 crypto assets

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SETUP Daily candles · frozen Kronos · day-7 backtests

BASELINE M1 UP Sharpe: −2.5744

UPDATED 29 October 2026 · 48 / 48 fits reviewed

STUDY SETUP

Experiment Dates

Daily candles · 7-day horizon · dates in YYYY-MM-DD.

Configuration	Data start	Train ends	Validation ends	Data ends
Earlier trial	2024-07-01	2025-05-30	2025-10-01	2026-01-25
Current · Pablo GPU8	2024-07-01	2025-08-31	2025-12-31	2026-08-01

Actual sample dates

Current experiment
Train: 2024-06-30 – 2025-08-31
Val-Cal: 2025-09-08 – 2025-10-18
Val-Opt: 2025-10-26 – 2025-12-31
Test: 2026-01-08 – 2026-07-18

Other evidence
Label diagnostics: 2024-07-01 – 2025-08-24
Earlier AG / Kumo Test:
2025-10-09 – 2026-01-18
Separate LightGBM Test:
2026-01-01 – 2026-09-23

Test dates match Pablo's reports. The original loader starts training one day before the configured start.

GPU8 config checked 2026-10-27; daily inputs checked 2026-10-28. Original context and embargo rules preserved. Earlier results have a separate PDF.

VERIFIED AGAINST GPU8

M1 Baseline

Test: 2026-01-08 – 2026-07-18 · spread equity · 7-day horizon · fee 0.2%.

M1 side	Return	Sharpe	Max drawdown	Trades
UP	−54.25%	**−2.5744**	−57.99%	1045
DOWN	30.44%	1.7423	−17.04%	2580

12 / 12
Baseline checks passed

27
Weekly returns
192 daily bars

−39.86%
Buy-and-hold return

UP loses money; DOWN gains. All six targets share the same M1 Test trades.

Saved GPU8 M1 metrics reproduced within 0.0001. Spread equity smooths returns; Sharpe annualization ×√(365.25/7). Exploratory Test.

THE "−45%" FIGURE

Loss and Drawdown

1045 UP trades · 2026-01-08 – 2026-07-18.

Equity calculation	Total return	Max drawdown	Sharpe
Spread · original baseline	−54.25%	−57.99%	−2.57
Cash · open positions at cost	−43.60%	−45.35%	−2.79
Cash · current market prices	−43.66%	−48.75%	−2.14

Return vs drawdown

−43.60%: total loss at the endpoint.
−45.35%: largest fall from a prior peak.

Valuation matters

Marking open positions to market reveals a larger drawdown: **−48.75%**.

Saved UP trades replayed 2026-10-28. Cash exposure: 10%; endpoint equity can include open positions. Compare results using the same valuation rule.

GPU8 REFERENCE

Pablo’s Saved Results

Test entries: 2026-01-08 – 2026-07-18 · 1d · TP labels · spread-equity engine.

Strategy	UP return	UP Sharpe	DOWN return	DOWN Sharpe
M1 · all calls	−54.25%	**−2.57**	30.44%	1.74
AutoGluon	−54.22%	**−2.36**	22.36%	1.58
CTTS	−58.65%	**−2.23**	23.36%	1.45
TabPFN	−21.50%	**−1.07**	20.70%	1.70
TabICL	−57.91%	**−2.43**	35.95%	1.91

Saved UP scores are negative. Our new comparison uses a fresh TP fit for each backend.

48 saved metrics replayed within 0.0001. M1, dates and engine match; new M2 fits on Izar differ from Pablo’s saved classifiers.

From Prices to Labels

Past prices
Frozen M1 proposes UP or DOWN.

→

Future prices
Apply exits and fees.

→

Meta-label z
1 if profitable; 0 otherwise.

Inspect labels

Compare exits and label changes on historical M1 calls. No retraining needed.

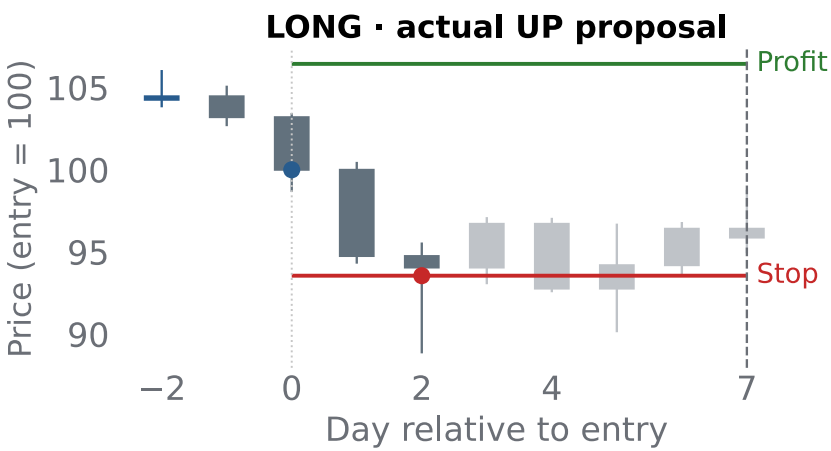
Test performance

Retrain M2 on each target, then compare. Keep M1 frozen.

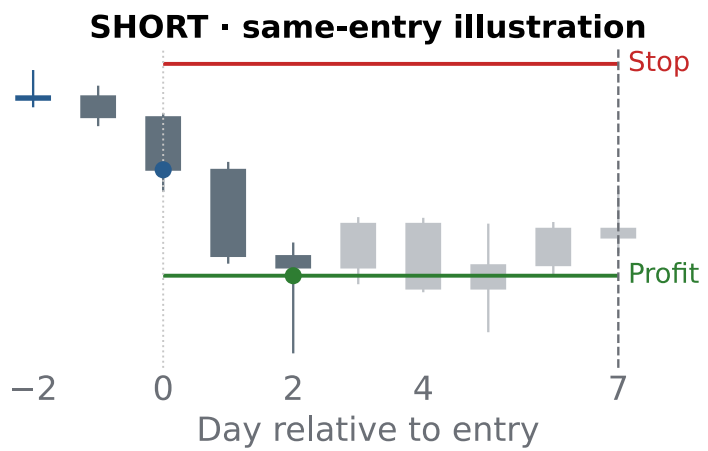
AFML §§3.4–3.7. Entry dates: 2024-07-01 – 2025-08-24. 7,593 proposals; 19/20 assets (TON unavailable).

Long and Short Barriers

BTC · 2024-07-03 · entry at day-0 close · symmetric k = 1 barriers.



Exit day 2 · net -6.6% · meta-label 0



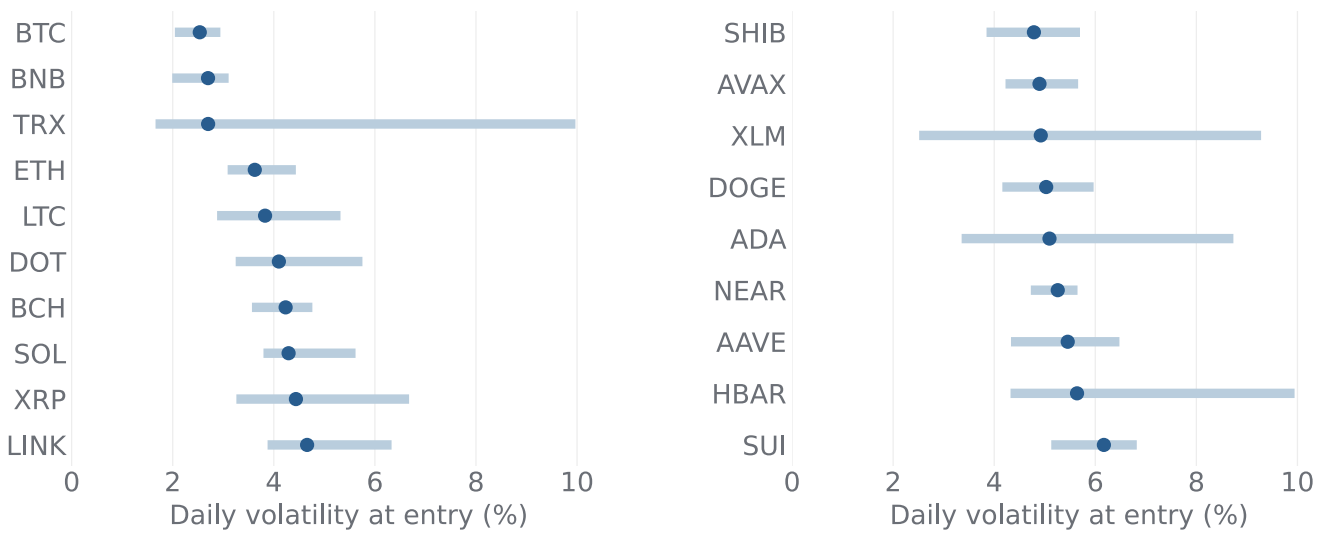
Exit day 2 · net +6.2% · meta-label 1

Long: profit above entry. **Short**: profit below entry. Stops are on the opposite side.

Training example selected for a stop; SHORT is counterfactual. Same volatility for both sides; 0.2% cost; assumed barrier fills.

Barrier Width by Asset

$$w_{a,t} = k\sigma_{a,t}\sqrt{7}$$

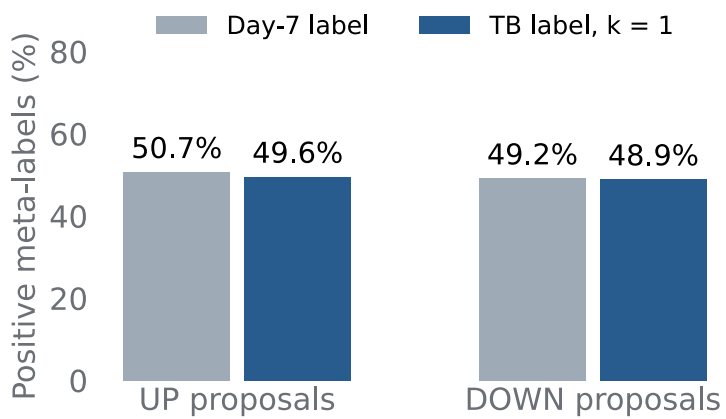


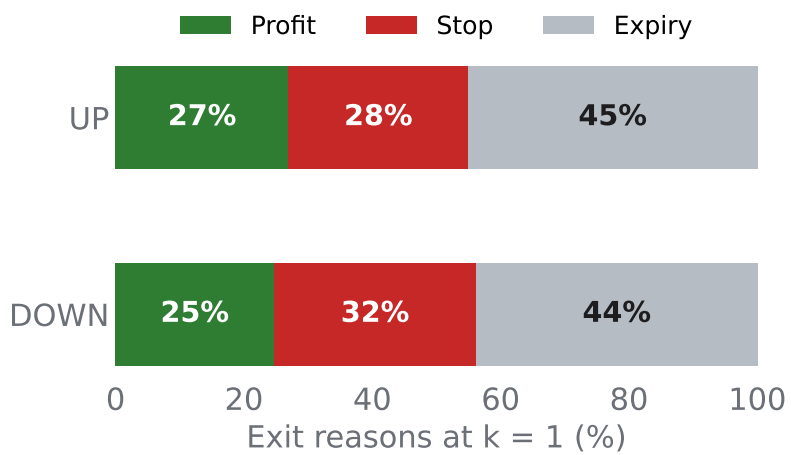
Same k, different widths: **BTC 6.7% · ETH 9.6% · SUI 16.3%** at k = 1.

Entry dates: 2024-07-01 – 2025-08-24. EWMA(100) at entry; width stays fixed per trade. Dots: median; lines: 10th–90th percentiles.

UP and DOWN Outcomes

1,921 UP proposals · 5,672 DOWN proposals · k = 1.



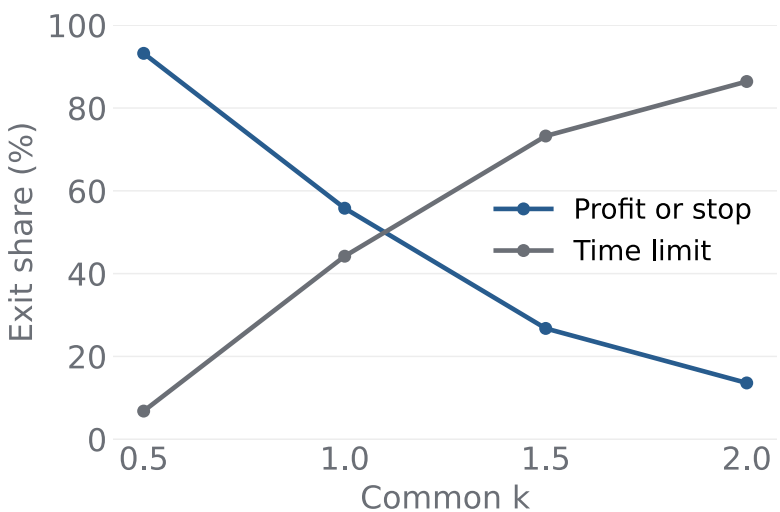


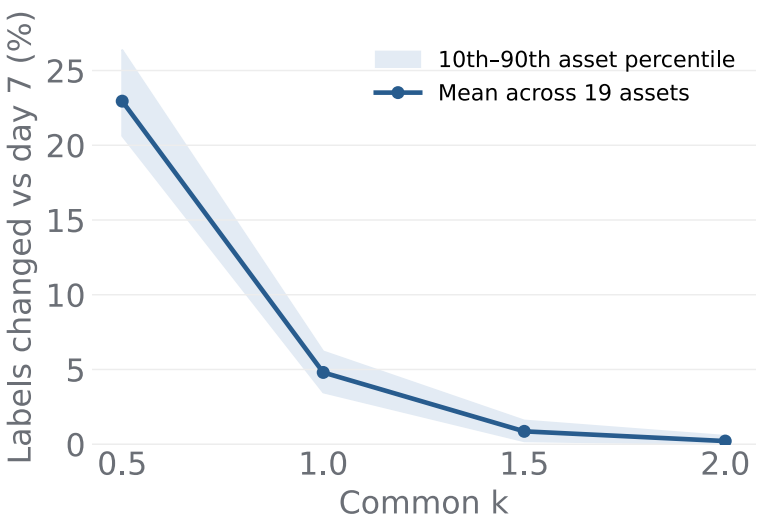
Labels change for **5.9% of UP** and **4.4% of DOWN** proposals.

Entry dates: 2024-07-01 – 2025-08-24. Different dates per side; descriptive comparison. High/low touches; same-candle ties = stop; cost 0.2%.

Sensitivity to k

Small k: closer exits. Large k: more trades remain open until day 7.





At k = 1: **55.8% profit/stop exits**, yet only **4.8% of labels change**.

Entry dates: 2024-07-01 – 2025-08-24. Equal asset weights; shading: 10th–90th percentiles. Training diagnostics do not select an optimal k.

Choosing k

One shared k

- All assets and both directions.
- Test 0.5, 1, 1.5, 2.
- Fix features, horizon and fees.

Validate the choice

- Define exits and the objective first.
- Check return, drawdown and stability.
- Confirm on fresh data.

Volatility already adapts the width. Add more parameters only with a stable validation gain.

Proposed protocol: AFML ch. 7 and 11. The 2026 Test has already been inspected; label balance alone is not an objective.

TP and TB Labels

TP = true-positive meta-labeling. Did the proposal profit at day 7 after the 0.2% fee?

Label 1

Positive net return at day 7.

Label 0

Zero or negative net return at day 7.

Column	M2 training target	Purpose
TP	Original day-7 label	Baseline
TB ref	TP label on TB-eligible samples	Control for the sample filter
TB k	Net outcome at profit / stop / expiry	Test a different label

TP does not mean take profit. All backtests keep day-7 exits.

TP rule verified on 18,230 complete proposals. Labels use future outcomes; M2 inputs use only information available at entry.

FOUR BACKENDS · 48 / 48 FITS REVIEWED

Test Sharpe

Test: 2026-01-08 – 2026-07-18 · day-7 exits.
M1 Sharpe: **UP −2.5744** · DOWN 1.7423. RF = Random Forest.

M2 fit	TP	TB ref	k = 0.5	k = 1	k = 1.5	k = 2
AutoGluon · UP	−2.199	−2.375*	−2.246	−2.376*	−2.305	−2.280
AutoGluon · DOWN	0.339	1.452	1.822	1.357	1.383	1.080
Kumo · UP	−1.983	−2.255	−2.463	−2.714	−2.038	−2.225
Kumo · DOWN	1.884	2.001	−0.278	1.731	1.849	1.730
TabPFN · UP	−2.527	−2.570	−2.353*	−2.594	−2.445	−2.519
TabPFN · DOWN	0.882	0.988	1.569	1.046	0.829	0.797
RF · UP	−1.800	−1.517	−2.222	−1.779	−1.388	−1.810
RF · DOWN	1.201	1.246	−0.473	−0.256	1.222	0.825

All UP scores are negative. Highest DOWN Sharpe: **Kumo TB ref, 2.001**.

Annualized spread-equity Sharpe; 27 weekly returns. * = utility constraint failed; threshold defaults to 0.5. One fit per cell; exploratory results.

SAME BACKEND AND DIRECTION

Change vs TP

Δ Sharpe = target − TP · positive values mean improvement.

M2 fit	TB ref	k = 0.5	k = 1	k = 1.5	k = 2
AutoGluon · UP	−0.176	−0.047	−0.178	−0.106	−0.081
AutoGluon · DOWN	+1.114	+1.483	+1.019	+1.044	+0.742
Kumo · UP	−0.272	−0.480	−0.731	−0.055	−0.242
Kumo · DOWN	+0.117	−2.162	−0.153	−0.035	−0.154
TabPFN · UP	−0.043	+0.174	−0.067	+0.082	+0.008
TabPFN · DOWN	+0.106	+0.687	+0.164	−0.054	−0.085
RF · UP	+0.283	−0.422	+0.021	+0.412	−0.010
RF · DOWN	+0.045	−1.675	−1.458	+0.020	−0.377

DOWN at k = 0.5: **AutoGluon +1.483** and **TabPFN +0.687** versus TP.

Includes the sample filter: TB removes **13 UP / 64 DOWN** early TON training proposals.

Test 2026-01-08–2026-07-18. Fresh TP controls; same Test trades, fees, features and day-7 exits. Descriptive deltas; no optimal-k claim.

SAME ELIGIBLE TRAINING SAMPLES

Change vs TB Reference

Δ Sharpe = TB-k − TB ref · compare labels on the same samples.

M2 fit	k = 0.5	k = 1	k = 1.5	k = 2
AutoGluon · UP	+0.128	−0.002	+0.069	+0.094
AutoGluon · DOWN	+0.370	−0.095	−0.069	−0.372
Kumo · UP	−0.208	−0.459	+0.217	+0.030
Kumo · DOWN	−2.279	−0.270	−0.152	−0.271
TabPFN · UP	+0.217	−0.024	+0.125	+0.051
TabPFN · DOWN	+0.580	+0.058	−0.160	−0.191
RF · UP	−0.704	−0.261	+0.130	−0.293
RF · DOWN	−1.719	−1.502	−0.024	−0.421

TabPFN DOWN · k = 0.5

Sharpe **1.569** · gain **+0.580**.
Still below M1 DOWN: **1.742**.

RF UP · k = 1.5

Sharpe **−1.388** · gain **+0.130**.
Return **−34.61%**: still loses money.

Test 2026-01-08–2026-07-18. Day-7 exits; 27 weekly returns; one fit per cell. Gains vary by backend and direction; no significance or optimal-k claim.